

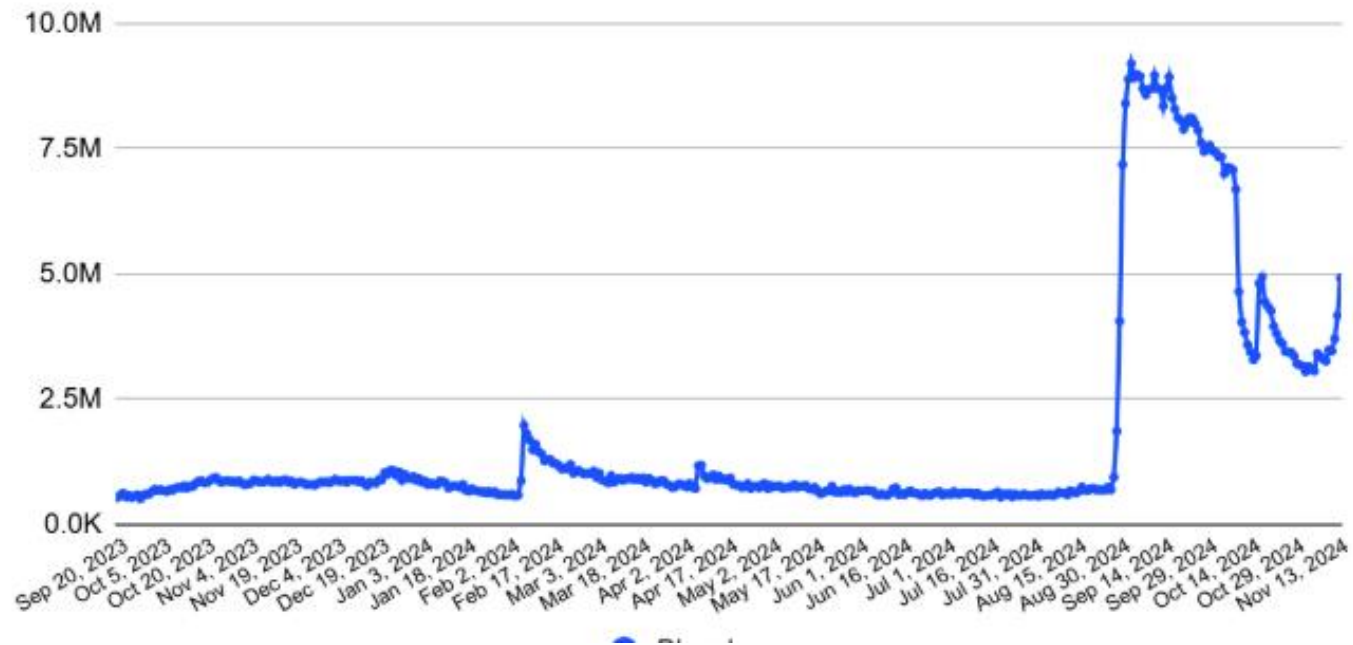
TYING EVERYTHING TOGETHER - THE THREE-TIER ARCHITECTURE

Week 9 Lecture
Joe Appleton
CSEE



Bluesky

Android Daily Active Users, Worldwide



IN THE NEWS

LAST WEEK WE



- Considered the role of a web framework in the development of a web application
- Considered the role of a web server in the development of a web application
- Used the Express framework to create a web application
- Use middleware in a web application
- Understand how to use `ejs` templates to create dynamic web pages
- Used `npm` to manage all of this

A CODE RECAP



THIS WEEK



- This week, we will conclude the web part of the module.
- Specifically, we will consider:
 - ✓ How to Create a full-stack web application using Node.js, Express, and **MySQL**.
 - ✓ The various compliance issues and best practices surrounding the development of web applications.
 - ✓ How to implement simple data validation on the server and client side.
 - ✓ Understand the three-tier architecture.

DISCLAIMER ABOUT THE LAB



- This week, I don't expect you to complete the lab in one sitting.
- I will release a full, in-depth tutorial video on how to complete the lab.
- After completing the lab, you'll be able to finish all non-advanced tasks of the assessment.

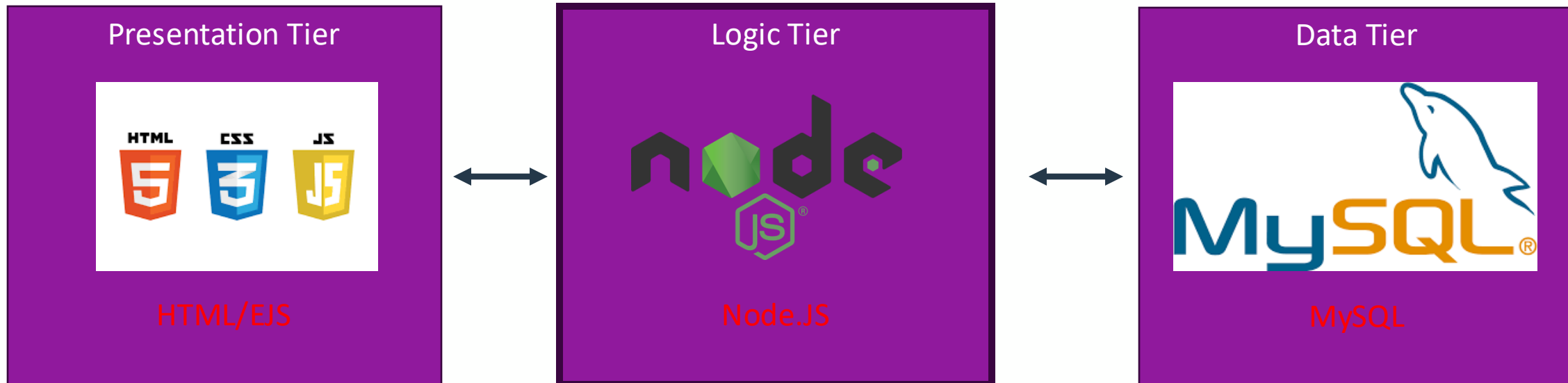
THE THREE-TIER ARCHITECTURE



WHAT IS A THREE TIER ARCHITECTURE?



- The three-tier architecture is a common architecture for web applications. It is a client-server architecture that separates the presentation layer, the application layer and the data layer.
- Improved scalability and maintainability.
- Clear separation of concerns.



THE PRESENTATION LAYER (CLIENT)



- is the layer that the user interacts with.
- responsible for rendering the user interface and handling user input.
- typically implemented using HTML, CSS and JavaScript.
- can create sophisticated presentation layers
- web Browsers: Browsers like Chrome, Firefox, and Safari are examples of clients that render and display web applications.
- mobile apps running on devices such as smartphones or tablets also serve as clients interacting with the presentation layer.



APPLICATION LAYER (SERVER)



- Business logic and application processing
 - authentication,
 - data validation
 - business rules
- Communication between presentation and data layers.
- Examples: Node.js, Django, Flask.

DATA LAYER (DATABASE)



- manages the storage of structured data in a database system.
- ensures efficient and secure access to stored information.
- handle tasks such as data organisation, indexing, and ensuring data integrity.
- examples: MySQL, PostgreSQL, MongoDB.

POSSIBLE DRAWBACKS OF A 3-TIER ARCHITECTURE



- Single point of failure.
- The communication between layers involves passing data back and forth. This can lead to increased overhead, especially if not managed efficiently.
- Developing a three-tier architecture can sometimes take more time compared to a simple HTML/CSS application.
- Changes in one layer may require corresponding adjustments in other layers

WIDER ARCHITECTURES



- **Microservices Architecture** involves breaking down a monolithic application into smaller, independent services that can be developed, deployed, and scaled independently.
- **Serverless Architecture** based on the concept of functions as a service (FaaS), where functions are triggered by events. These events can be HTTP requests, database changes, file uploads, etc.

CONNECTING TO A DATABASE



FIRST, WE NEED A DATABASE TO CONNECT TO AND NODEJS INSTALLED



Downloads

Latest LTS Version: 20.10.0 (includes npm 10.2.3)

Download the Node.js source code or a pre-built installer for your platform, and start developing today.

LTS
Recommended For Most Users

Current
Latest Features


Windows Installer
node-v20.10.0-x64.msi


macOS Installer
node-v20.10.0.pkg


Source Code
node-v20.10.0.tar.gz

Windows Installer (.msi)	32-bit	64-bit	ARM64
Windows Binary (.zip)	32-bit	64-bit	ARM64
macOS Installer (.pkg)	64-bit / ARM64		
macOS Binary (.tar.gz)	64-bit	ARM64	
Linux Binaries (x64)	64-bit		
Linux Binaries (ARM)	ARMv7	ARMv8	
Source Code	node-v20.10.0.tar.gz		

Additional Platforms

Docker Image	Official Node.js Docker Image
Linux on Power LE Systems	64-bit
Linux on System z	64-bit
AIX on Power Systems	64-bit

<https://nodejs.org/en/download>

General Availability (GA) Releases

Archives



MySQL Installer 8.0.35



Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

8.0.35

Select Operating System:

Microsoft Windows

Windows (x86, 32-bit), MSI Installer

8.0.35

2.1M

[Download](#)

(mysql-installer-web-community-8.0.35.0.msi)

MD5: 214df2ccdf83eb5edc6ca7c115792406 | [Signature](#)

Windows (x86, 32-bit), MSI Installer

8.0.35

288.6M

[Download](#)

(mysql-installer-community-8.0.35.0.msi)

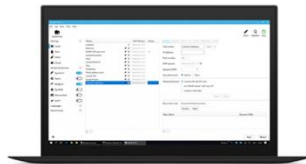
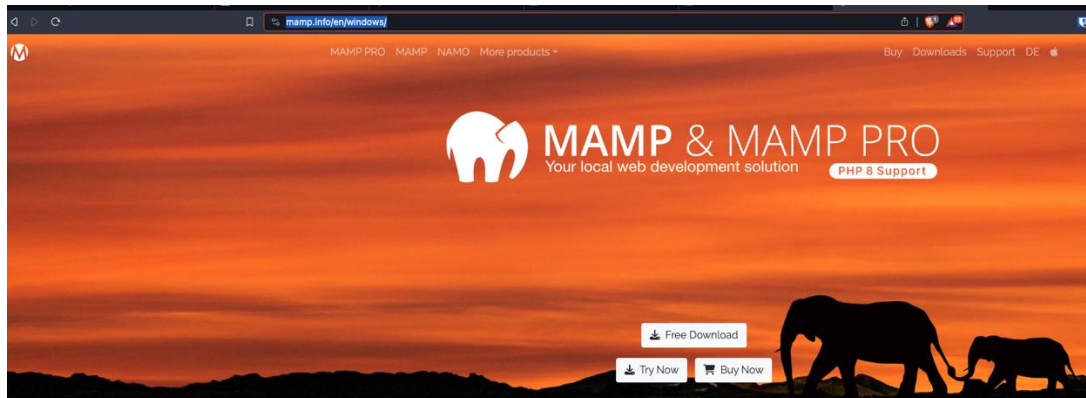
MD5: 2cfd448a2971b6b5323775ef9e8d012 | [Signature](#)



We suggest that you use the [MD5 checksums](#) and [GnuPG signatures](#) to verify the integrity of the packages you download.

<https://dev.mysql.com/downloads/mysql/>

YOU CAN ALSO INSTALL A BUNDLED MYSQL AND NODE PACKAGE

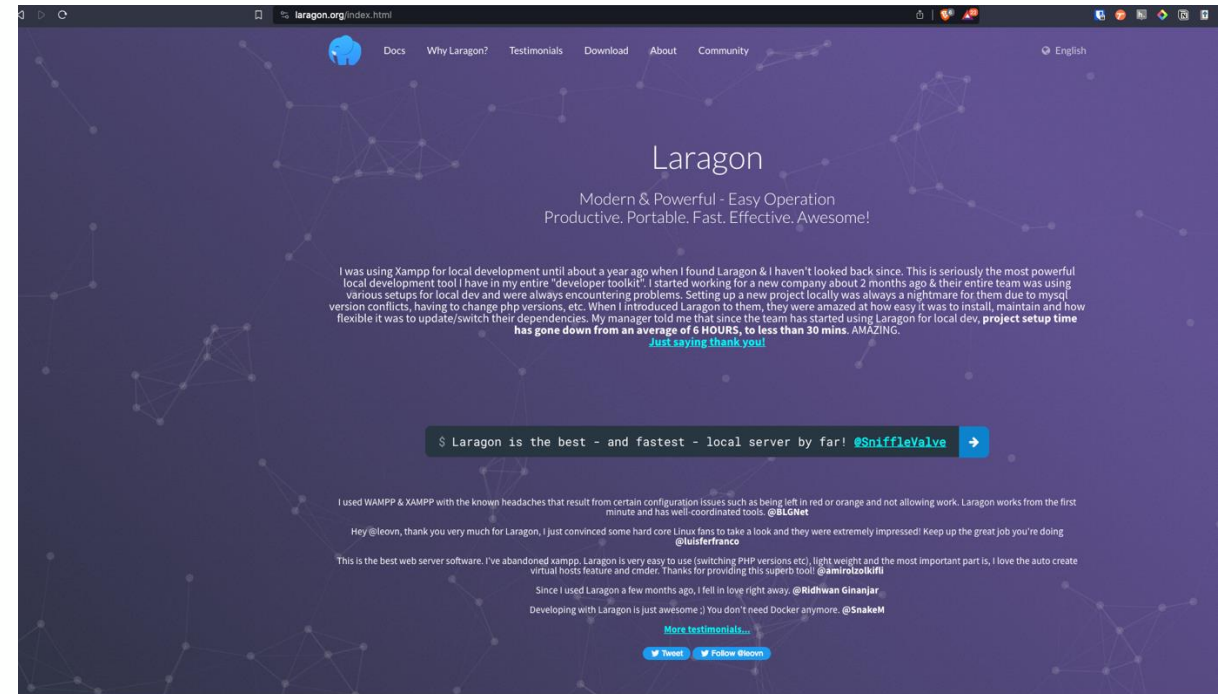


MAMP PRO for Windows

MAMP PRO is the commercial, **professional frontend** for the classic local server environment: MAMP. With MAMP PRO you can create a separate host for each of your web projects. You can install **WordPress** and other Extras like Joomla or Drupal with just a few clicks. This is just a small part of what you can do with MAMP PRO.

[Learn more about MAMP PRO >](#)

[Buy MAMP PRO for Windows >](#)





CONNECTING TO A DATABASE CODE

ONCE CONNECTED, WE CAN RUN MYSQL QUERIES



```
const studentCount = await connection.query("SELECT COUNT(*) as count  
FROM Student");
```

- The above code runs the query "SELECT COUNT(*)as count FROM Student" and stores the result in a constant called studentCount.
- Notice how we use the **await** keyword. This is because the connection.query function is asynchronous.
- studentCount will be an array of objects. In this case, it will be an array with one object:



```
[  
  {  
    count: 10,  
  },  
]
```

```
console.log(studentCount[0].count) // logs 10 to the console
```

QUERIES THAT RETURN MULTIPLE ROWS



```
const students = await connection.query("SELECT * FROM Student");
```

QUERIES THAT RETURN MULTIPLE ROWS



```
const students = await connection.query("SELECT * FROM Student");
```

[

```
RowDataPacket {  
  URN: 612345,  
  Stu_FName: 'John',  
  Stu_LName: 'Smith',  
  Stu_DOB: 2002-06-19T23:00:00.000Z,  
  Stu_Gender: 'M',  
  Stu_Phone: '11111',  
  Stu_Course: 201,  
  Stu_Type: 'UG'  
},
```

```
RowDataPacket {  
  URN: 612345,  
  Stu_FName: 'John',  
  Stu_LName: 'Smith',  
  Stu_DOB: 2002-06-19T23:00:00.000Z,  
  Stu_Gender: 'M',  
  Stu_Phone: '11111',  
  Stu_Course: 201,  
  Stu_Type: 'UG'  
},
```

```
},
```

.....



ACCESSING VALUES IN AN ARRAY



```
students.forEach(student => { console.log(student) });
```

THE PRESENTATION LAYER



HTML AND CSS



- HTML and CSS form a powerful duo for building the presentation layer of web applications
- HTML: Provides a lightweight structure for content,
- CSS: Centralising styles in separate CSS files makes it easier to maintain and update the design of a website.
- simple to learn
- if you have not gone over my pre-learning, ensure you do this.

WHAT'S WRONG WITH JUST HTML



- Serving plain HTML files provides us with a means to present a website
- How do we inject data into our html pages?
- Shared components or layouts must be replicated across different HTML files
- As web applications grow in complexity, maintaining a large codebase written in pure HTML becomes increasingly challenging.
- The lack of a clear separation of concerns makes it difficult to manage the presentation layer independently from the application logic

EJS: TEMPLATING TO THE RESCUE



- Templating allows for the dynamic generation of content in web applications
- Templating promotes code reusability by separating the HTML structure from the data
- Shared components
- Templating ensures consistent layouts across different pages of a website.

EJS ALLOWS HTML TO BE CHUNKED



```
<!-- views/common/header.ejs -->
<ul>
  <li><a href="home">Home</a></li>
  <li><a href="about"></a>About</li>
  <li><a href="contact">Contact</a></li>
</ul>
```

```
<!-- views/index.ejs -->
...
<body>
  <%- include('common/header'); %>
  <h1>Home Page</h1>
  
</body>
...
```

YOU CAN ALSO INJECT DATA



```
// index.js
app.get('/', async (req, res) => {
  res.render('index',
    {
      name: "joe appleton",
      location: "surrey"
    }
  );
});
```

```
<!-- views/index.ejs -->
...
<h1> Hello World, <%= name %> </h1> <!-- Hello World, joe appleton-->
<p> You are currently in <%= location %> </p> <!-- You are currently in , brighton -->
...
```

YOU CAN ALSO INJECT DATA



```
// index.js
app.get('/', async (req, res) => {
  res.render('index', { students: [

    { name: "Joe Appleton", location: "Surrey" },
    { name: "Jane Doe", location: "London" },
    { name: "John Smith", location: "Manchester" },
  ]}):

});
```

```
<ul>

  <% students.forEach(function(student){ %>
    <li>
      <%= student.name %> - <%= student.location %>
    </li>
  <% }) %>

</ul>
```

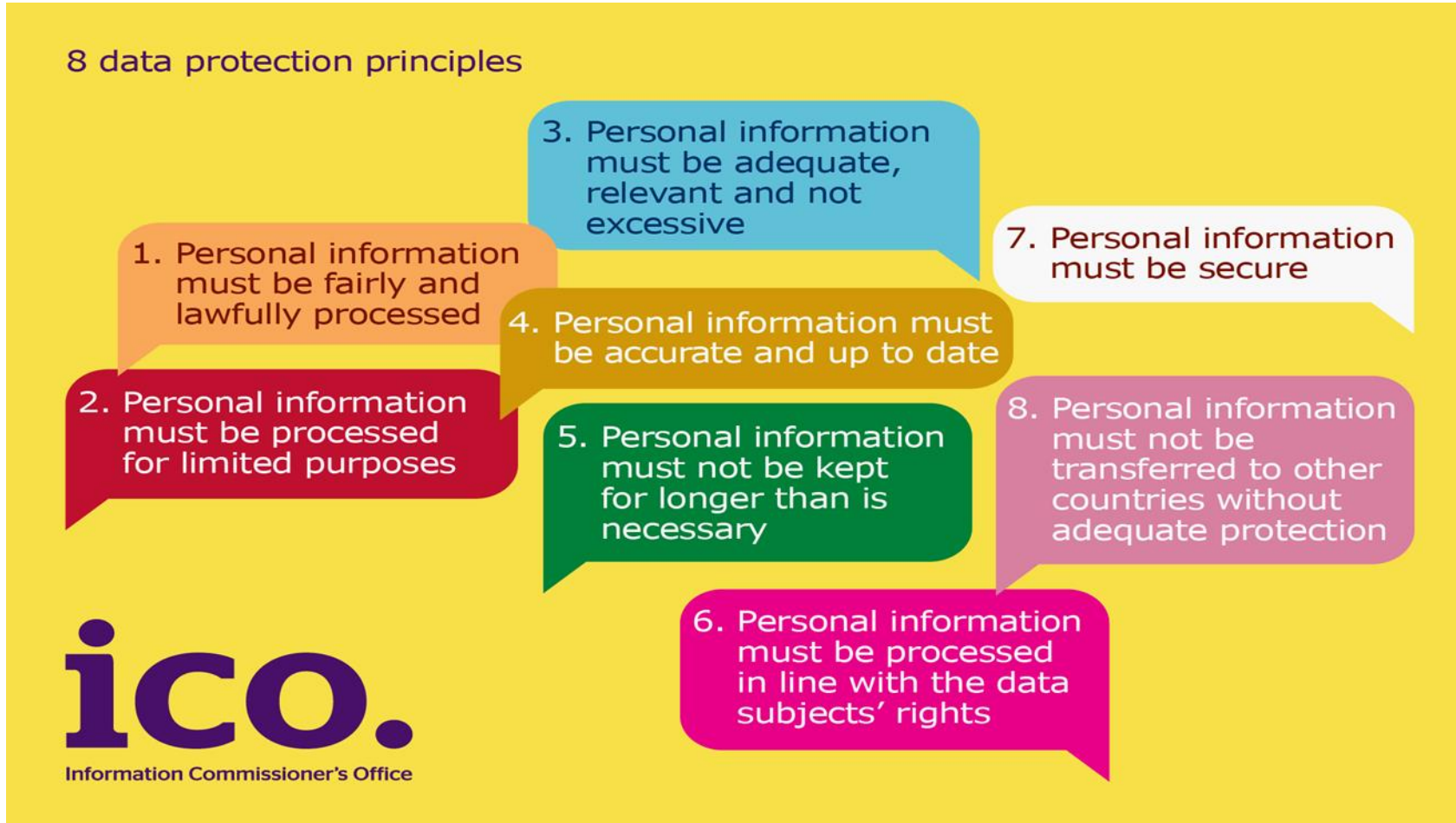
CONCERNS WHEN DEALING WITH USER DATA



LEGAL



Data Protection Act 1998 The eight principles



| British Pregnancy Advice Service

Date: 7 March 2014

**British Pregnancy Advice Service
fined £200,000. Hacker
threatened to publish thousands
of names of people who sought
advice on abortion, pregnancy
and contraception.**



**Abortion provider BPAS
fined £200,000 for data
breach**



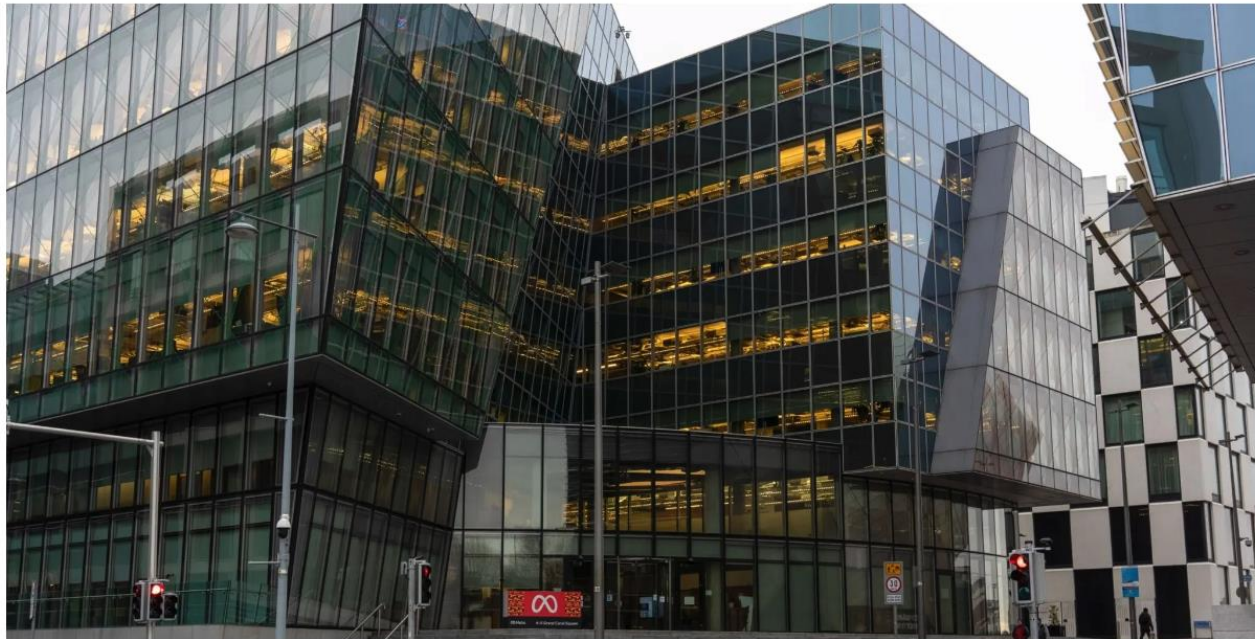
Meta Fined \$1.3 Billion for Violating E.U. Data Privacy Rules

The Facebook owner said it would appeal an order to stop sending data about European Union users to the United States.

 Share full article



 234



Meta settles Cambridge Analytica scandal case for \$725m

🕒 23 December 2022



Facebook-Cambridge Analytica scandal



GETTY IMAGES

By Shiona McCallum

Technology reporter

Facebook owner Meta has agreed to pay \$725m (£600m) to settle legal action over a data breach linked to political consultancy Cambridge Analytica.

The long-running dispute accused the social media giant of allowing third parties, including the British firm, to access Facebook users' personal data

THE GENERAL DATA PROTECTION REGULATION (GDPR)



Lawfulness, Fairness, and Transparency

Purpose Limitation

Data Minimisation

Accuracy

Integrity and Confidentiality (Security)

Accountability

TECHNICAL



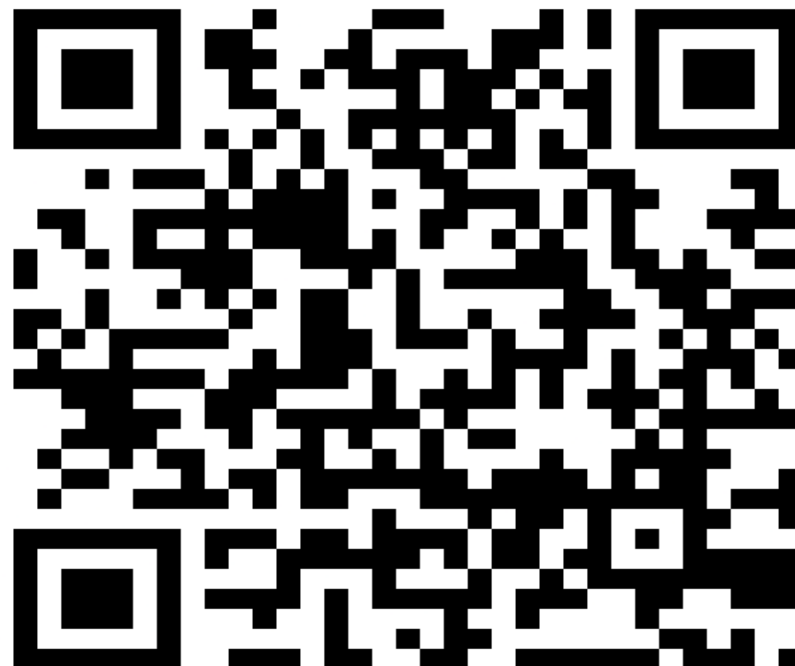
SANITISING USER INPUT



NOT SANITIZING USER INPUT CAN LEAD TO VARIOUS SECURITY VULNERABILITIES



- ✓ SQL Injection:
- ✓ Code Injection
- ✓ Command Injection



Visit: <https://http-server-leqm.onrender.com/>



ESCAPE EJS TEMPLATES



```
<%= userName %> <!-- this sanitises the output -->
```

```
<%- userName %> <!-- this this does not sanitise the output -->
```

USE PARAMETERIZED QUERY TO PREVENT SQL INJECTION



```
app.get('/students/view/:id', async (req, res) => {  
    const student = await connection.query('SELECT * FROM STUDENT WHERE  
    URN = ?', [req.params.id]);  
    res.render('student_view', { student: student[0] });  
});
```

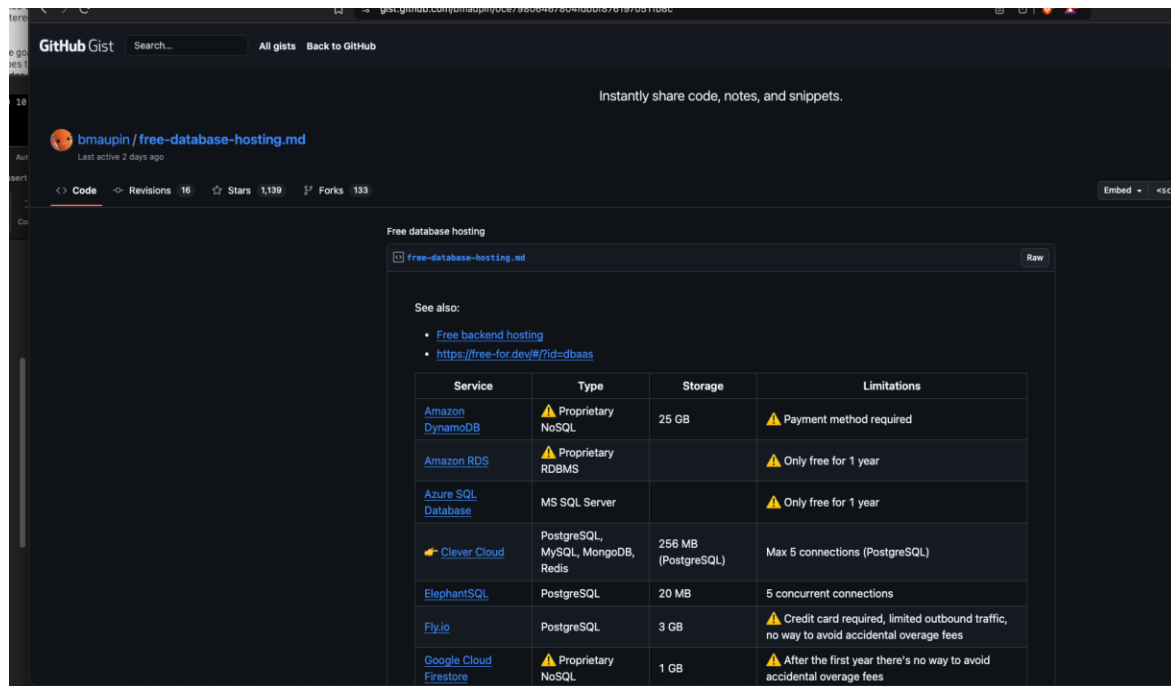
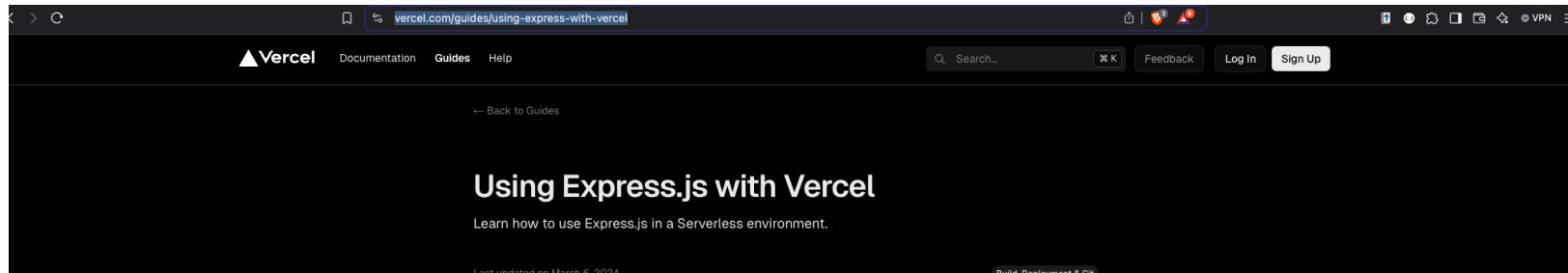
DEPLOYMENT



SERVERLESS DEPLOYMENT



<https://vercel.com/guides/using-express-with-vercel>



<https://gist.github.com/bmaupin/0ce79806467804fdbbf8761970511b8c>

END OF LECTURE

